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Assignment Part A

02435 Decision Making under Uncertainty

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1 Introduction

The scope of the problem is to design the operation of a restaurant's heating and ventilation system. The entire restaurant has two rooms. Each room is equipped with an electric heater that modulates its heating power and sensors that measure indoor temperature and humidity. In addition to that, the restaurant is equipped with a central ventilation unit. The temperature in each room is dependent on the temperature in the other room, the outdoor temperature, the power at which the heater runs, whether the ventilation is ON or OFF, and the occupancy level. Meanwhile, the humidity level is dependent on the occupancy and the ventilation status. The system should be designed in the most efficient way, cost-wise.

2 Task 1: Optimal in Hindsight solution

The condition for this task is to formulate and implement the Optimal in Hindsight solution as a mixed-integer linear program (MILP). That is done with the condition that all uncertainties, like the occupancy level of each room and the electricity price, are perfectly known beforehand. This implies that at the beginning of each day, the system makes smart decisions like turning on the ventilation when the price is low, although the humidity level is within the normal range, just because it knows that the occupancy level is high later on when prices are high, making it cheaper to ventilate in advance. In practice, this perfect foresight assumption would not hold, as prices and occupancy are uncertain at the time of decision. The optimal in hindsight solution, therefore, serves as a lower bound on the achievable cost under any real-time control policy.

2.1 MILP formulation

Sets

- $\mathcal{T}$: Set of time slots, $t \in \mathcal{T}$
- $\mathcal{R}$: Set of the restaurant rooms, $r \in \mathcal{R}$

Decision variables

- $p_{r,t}$: Power of the heater in room r at time t , $0 \leq p_{r,t} \leq \bar{P}_r$
- v_t : Binary variable indicating whether the ventilation is on or off at time t , $v_t \in \{0, 1\}$
- $Ltemp_{r,t}$: Binary variable indicating whether the low temperature overrule controller is on or off in room r at time t , $Ltemp_{r,t} \in \{0, 1\}$
- $Htemp_{r,t}$: Binary variable indicating whether the high temperature overrule controller is on or off in room r at time t , $Htemp_{r,t} \in \{0, 1\}$
- $temp_{r,t}$: Temperature in room r at time t

- hum_t : Humidity at time t
- $z_{r,t}^{trig}$: Binary variable indicating whether the temperature in room r at time t is below T^{low} threshold
- $z_{r,t}^{rec}$: Binary variable indicating whether the temperature in room r at time t is above the T^{OK} threshold

Parameters

- λ_t : electricity price at time t
- $\bar{P}_r$: Maximum power at which the electric heater can operate
- P^{vent} : Power of the ventilator when turned on
- T^{low} : Low temperature threshold
- T^{OK} : Recovery temperature threshold
- T^{High} : High temperature threshold
- H^{high} : High humidity threshold
- $occ_{r,t}$: Occupancy level in room r and time t
- ζ^{exch} : Room temperature exchange coefficient
- ζ^{loss} : Heat loss coefficient
- ζ^{conv} : Room heat gain coefficient
- ζ^{cool} : Room ventilation temperature coefficient
- ζ^{occ} : Room occupancy temperature coefficient
- η^{occ} : Room occupancy humidity coefficient
- η^{vent} : Room ventilation humidity coefficient
- T_t^{out} : Outdoor temperature at time t
- T_r^{init} : Initial temperature in room r
- H^{init} : Initial humidity level

Objective Function

$$\min \sum_{t \in \mathcal{T}} \sum_{r \in \mathcal{R}} p_{r,t} \cdot \lambda_t + \sum_{t \in \mathcal{T}} P^{vent} \cdot v_t \cdot \lambda_t \quad (1)$$

Constraints

Temperature dynamics: The temperature in each room evolves hour by hour based on heat exchange with the other room, heat loss to the outside, heat gained from the heater, cooling from ventilation, and heat generated by occupants.

$$\begin{aligned} temp_{r,t} = & temp_{r,t-1} + \zeta^{exh}(temp_{r',t-1} - temp_{r,t-1}) \\ & + \zeta^{loss}(T_{t-1}^{out} - temp_{r,t-1}) \\ & + \zeta^{conv} \cdot p_{r,t-1} \\ & - \zeta^{cool} \cdot v_{t-1} \\ & + \zeta^{occ} \cdot occ_{r,t-1}, \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \end{aligned} \quad (2)$$

$$temp_{r,0} = T_r^{init}, \quad \forall r \in \mathcal{R} \quad (3)$$

where $r' \in \mathcal{R} \setminus \{r\}$ denotes the other room.

Humidity dynamics: The humidity level evolves hour by hour, increasing with occupancy and decreasing when ventilation is active.

$$hum_t = hum_{t-1} + \eta^{occ} \cdot \sum_{r \in \mathcal{R}} occ_{r,t-1} - \eta^{vent} \cdot v_{t-1}, \quad \forall t \in \mathcal{T} \quad (4)$$

$$hum_0 = H^{init} \quad (5)$$

Temperature Overrule Controller: The idea behind the overrule is that certain thresholds trigger the heater power. When the temperature drops below T^{low} , the overrule activates, forcing the heater to operate at its maximum power. Once the temperature recovers to T^{OK} , the overrule deactivates, allowing the optimizer to choose any power in the range $[0, \bar{P}]$. When the temperature exceeds T^{high} , the overrule activates for that hour, forcing the heater off.

Low temperature:

Constraints (6) and (7) define the trigger binary $z_{r,t}^{trig}$, which equals 1 if and only if the room temperature falls at or below T^{low} at time t . Constraint (6) prevents the trigger from activating when the temperature is above the threshold, while constraint (7) forces the trigger to activate when the temperature drops below it. ($M = T^{High} - T^{low}$)

$$temp_{r,t} \leq T^{low} + M(1 - z_{r,t}^{trig}) \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \setminus \{0\} \quad (6)$$

$$temp_{r,t} \geq T^{low} - M \cdot z_{r,t}^{trig} \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \setminus \{0\} \quad (7)$$

Constraints (8) and (9) define the recovery binary $z_{r,t}^{rec}$, which equals 1 if and only if the room temperature reaches or exceeds T^{OK} at time t . Constraint (8) prevents the recovery

from activating when the temperature is still below the threshold, while constraint (9) forces the recovery to activate once the temperature reaches T^{OK} .

$$temp_{r,t} \geq T^{OK} - M(1 - z_{r,t}^{rec}) \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \setminus \{0\} \quad (8)$$

$$temp_{r,t} \leq T^{OK} + M \cdot z_{r,t}^{rec} \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \setminus \{0\} \quad (9)$$

Constraints (10)–(13) define the persistence logic. $Ltemp_{r,t}$. Constraint (10) ensures the controller activates whenever a new trigger fires. Constraint (11) enforces persistence, once active, the controller remains on until recovery is detected. Constraint (12) prevents the controller from activating without a trigger or prior activation. Finally, constraint (13) forces the controller to deactivate once the temperature recovers to T^{OK} , unless a new trigger fires simultaneously.

$$Ltemp_{r,t} \geq z_{r,t}^{trig} \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \quad (10)$$

$$Ltemp_{r,t} \geq Ltemp_{r,t-1} - z_{r,t}^{rec} \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \setminus \{0\} \quad (11)$$

$$Ltemp_{r,t} \leq z_{r,t}^{trig} + Ltemp_{r,t-1} \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \setminus \{0\} \quad (12)$$

$$Ltemp_{r,t} \leq 1 - z_{r,t}^{rec} + z_{r,t}^{trig} \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \quad (13)$$

When the controller is active ($Ltemp_{r,t} = 1$), the heater is forced to operate at its maximum power $\bar{P}$. When the controller is inactive ($Ltemp_{r,t} = 0$), the constraint becomes non-binding, and the optimizer is free to choose any power level in $[0, \bar{P}]$ to minimize cost.

$$p_{r,t} \geq \bar{P} - M(1 - Ltemp_{r,t}) \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \quad (14)$$

Constraint (15) defines the initial condition. Since no previous controller state exists, the overrule can only be active at the start of the horizon if the temperature is already below T^{low} . As the initial temperature is fixed at $T^{init} = 21^\circ\text{C}$, which lies within the comfort zone, the controller is always inactive at $t = 0$.

$$Ltemp_{r,0} = z_{r,0}^{trig} \quad \forall r \in \mathcal{R} \quad (15)$$

High temperature: When the controller is active ($Htemp_{r,t} = 1$), the temperature must be at or above T^{High} , triggering the overrule. When the controller is active, the heater is forced OFF, setting power to zero.

$$temp_{r,t} \geq T^{High} - M(1 - Htemp_{r,t}) \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \quad (16)$$

$$temp_{r,t} \leq T^{High} + M \cdot Htemp_{r,t} \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \quad (17)$$

$$p_{r,t} \leq M(1 - Htemp_{r,t}) \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \quad (18)$$

Mutual exclusion: It can not be too cold and hot simultaneously

$$Ltemp_{r,t} + Htemp_{r,t} \leq 1 \quad \forall r \in \mathcal{R}, t \in \mathcal{T} \quad (19)$$

Humidity Overrule Controller: When humidity exceeds H^{high} , the ventilation sys-

tem is forced ON to reduce it.

$$hum_t \leq H^{high} + M \cdot v_t, \quad \forall t \in \mathcal{T} \quad (20)$$

Ventilation System Inertia: Once the ventilation is switched ON, it must remain on for at least 3 consecutive hours.

$$v_t + v_{t+1} + v_{t+2} \geq 3(v_t - v_{t-1}) \quad \forall t \in \mathcal{T} \quad (21)$$

2.2 Python implementation

Link to the GitHub repository

Average daily electricity cost: 112.9059

2.3 Result interpretations

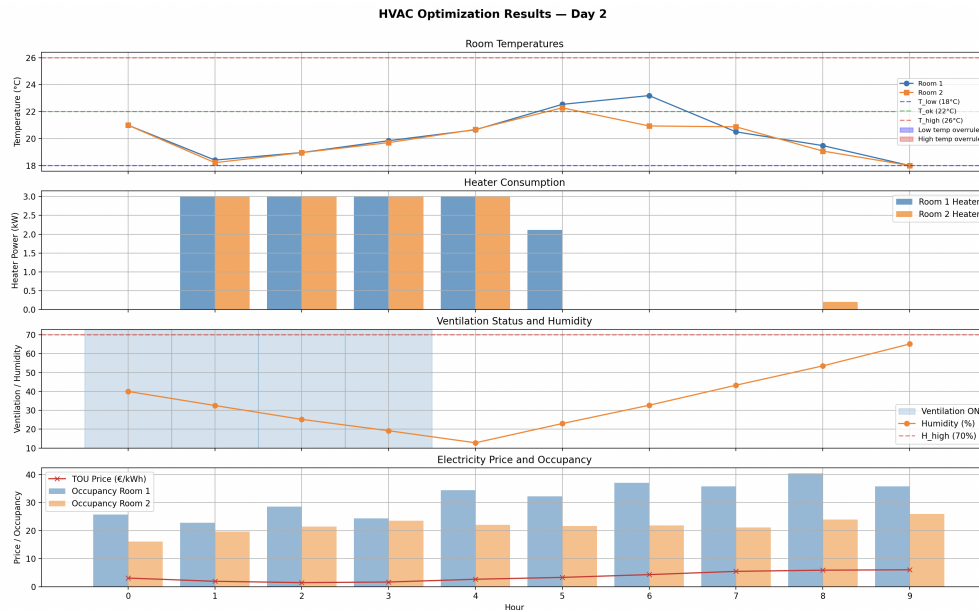


Figure 1: MILP result for day 2

The plot for day 2 indicates that ventilation is activated at $t = 0$, which is not due to high humidity, as its initial value is below 70%. This reflects the optimizer's perfect foresight, knowing that occupancy levels are high throughout the day, which drives humidity up, it deliberately ventilates when it's cheapest to do so, avoiding ventilating later when prices are high. The ventilation runs for more than 3 hours, which is as expected. When the ventilation is on, the optimizer sets the heater to maximum power to compensate for the decrease in temperature from the ventilation. The power is set at its maximum value due to low cost.

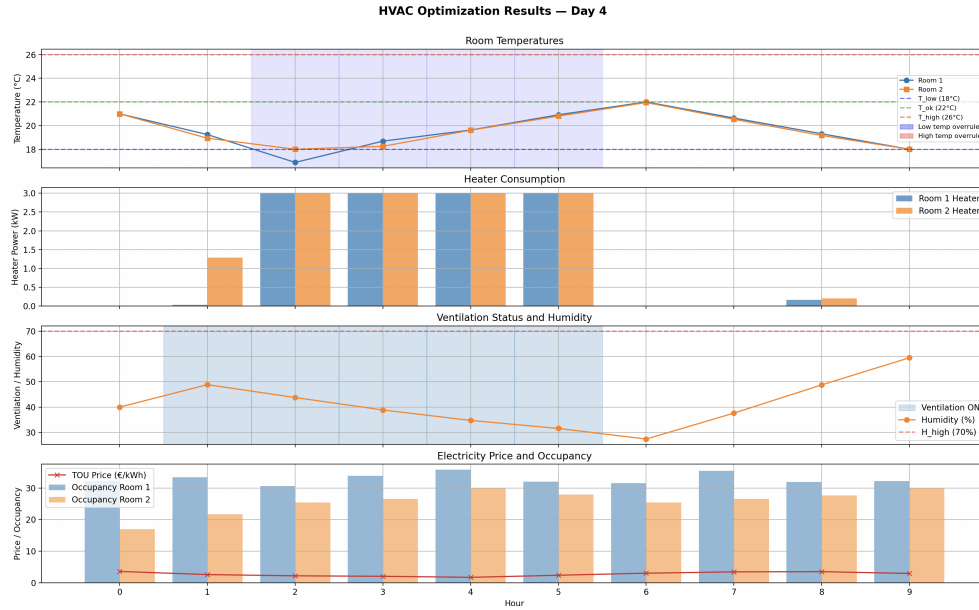


Figure 2: MILP result for day 4

The plot for day 4 indicates a temperature drop below the T^{low} threshold, which activates the controller and sets the heater to its maximum power. As soon as one of the rooms recovers to the T^{OK} temperature, the controller is turned off, proving the correct behavior modeled. The ventilation follows the same principle as on day 2.

3 Task 2: Markov Decision Process Formulation

The problem is formulated as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$.

1. State Space ($\mathcal{S}$)

To satisfy the Markov property, the state $s_t \in \mathcal{S}$, observed at the beginning of hour t (after exogenous variables for hour t are revealed), must encapsulate the current physical conditions, the history required to predict future uncertain exogenous variables, and the memory required by the control logic.

$$s_t = \left(\{T_{r,t}\}_{r \in \mathcal{R}}, H_t, \lambda_t, \lambda_{t-1}, \{O_{r,t}\}_{r \in \mathcal{R}}, v_{t-1}, v_{t-2}, \{\delta_{r,t}^{low}\}_{r \in \mathcal{R}} \right)$$

where:

- $T_{r,t} \in \mathbb{R}$: Indoor temperature of room $r \in \mathcal{R}$ at hour t
- $H_t \in \mathbb{R}$: Indoor humidity of the restaurant at hour t

- $\lambda_t, \lambda_{t-1} \in \mathbb{R}_+$: Electricity prices at hours t and $t - 1$. Since λ_{t+1} is a function of $(\lambda_t, \lambda_{t-1})$, both are needed to predict future prices. The current price λ_t is included as it enters the reward function directly.
- $O_{r,t} \in \mathbb{N}$: Occupancy level of room r at hour t . Since $O_{r,t+1}$ depends on $(O_{1,t}, O_{2,t})$, the current occupancies of both rooms are required.
- $v_{t-1}, v_{t-2} \in \{0, 1\}$: Ventilation status at hours $t - 1$ and $t - 2$. These two binary values are necessary and sufficient to enforce the 3-hour minimum uptime constraint: if $v_{t-2} = 0$ and $v_{t-1} = 1$, ventilation must remain ON at hours t and $t + 1$; if $v_{t-2} = 1$ and $v_{t-1} = 1$, it must remain ON at hour t only.
- $\delta_{r,t}^{low} \in \{0, 1\}$: Binary indicator that the low-temperature overrule controller is active in room r at hour t , forcing the heater to maximum power until $T_{r,t} \geq T^{OK}$. This cannot be inferred from $T_{r,t}$ alone (e.g., $T_{r,t} \in [T^{low}, T^{OK})$ is ambiguous without this flag).

2. Decision Variables ($\mathcal{A}$)

The action $a_t \in \mathcal{A}(s_t)$ taken at hour t is:

$$a_t = \left(\{p_{r,t}\}_{r \in \mathcal{R}}, v_t \right)$$

where:

- $p_{r,t} \in [0, \bar{P}_r]$: Heating power for room r (kW).
- $v_t \in \{0, 1\}$: Central ventilation unit status (1 = ON, 0 = OFF).

The feasible action space $\mathcal{A}(s_t) \subseteq \mathcal{A}$ is state-dependent due to the following constraints:

- If $\delta_{r,t}^{low} = 1$: $p_{r,t} = \bar{P}_r$ (heater locked at maximum by low-temperature overrule controller).
- If $T_{r,t} > T^{high}$: $p_{r,t} = 0$ (heater forced off by high-temperature overrule controller).
- If $H_t > H^{high}$: $v_t = 1$ (ventilation forced ON by humidity overrule controller).
- If $v_{t-2} = 0, v_{t-1} = 1$: $v_t = 1$ (ventilation locked ON due to minimum uptime inertia).

3. Reward Function ($\mathcal{R}$)

The objective is to minimize total electricity cost. The per-step reward is defined as the negative cost incurred at hour t :

$$R(s_t, a_t) = -\lambda_t \left(\sum_{r \in \mathcal{R}} p_{r,t} + P^{vent} \cdot v_t \right)$$

The overall objective is:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=1}^T R(s_t, \pi(s_t)) \right]$$

where $\pi : \mathcal{S} \rightarrow \mathcal{A}$ is the policy.

4. Transition Dynamics ($\mathcal{P}$)

The transition to s_{t+1} is deterministic for physical variables and stochastic for exogenous variables.

- **Temperature** (known, deterministic):

$$T_{r,t+1} = T_{r,t} + \zeta^{exch}(T_{r',t} - T_{r,t}) - \zeta^{loss}(T_{r,t} - T_t^{out}) + \zeta^{conv}p_{r,t} - \zeta^{cool}v_t + \zeta^{occ}O_{r,t}, \quad r' \neq r$$

- **Humidity** (known, deterministic):

$$H_{t+1} = H_t + \eta^{occ}(O_{1,t} + O_{2,t}) - \eta^{vent}v_t$$

- **Electricity price** (unknown, stochastic):

$$\lambda_{t+1} \sim f(\lambda_t, \lambda_{t-1}, \xi_t^\lambda)$$

where $f(\cdot)$ is an unknown function and ξ_t^λ is exogenous noise.

- **Occupancy** (unknown, stochastic):

$$O_{r,t+1} \sim g_r(O_{1,t}, O_{2,t}, \xi_{r,t}^{occ}), \quad r \in \mathcal{R}$$

where $g_r(\cdot)$ is an unknown function and $\xi_{r,t}^{occ}$ is exogenous noise.

- **Ventilation memory** (known, deterministic):

$$v_{t-1} \leftarrow v_t, \quad v_{t-2} \leftarrow v_{t-1}$$

- **Low-temperature overrule lock** (known, deterministic):

$$\delta_{r,t+1}^{low} = \begin{cases} 1, & \text{if } T_{r,t+1} < T^{low} \\ 0, & \text{if } T_{r,t+1} \geq T^{OK} \\ \delta_{r,t}^{low}, & \text{otherwise} \end{cases}$$

4 Contributions

The following contributions are assigned to the group members:

Simona Tican: I have entirely formulated and implemented task 1. The part of the report associated with this task and the introduction was written by me. Task 2 was only subject of review. I did not have a solution to cross-validate it with. Only read it through and considered it a good solution overall. This concludes my contribution of 40%.

Kushagra Kapoor: I contributed to the formulation of the Markov Decision Process (Task 2) by defining the state space, action space, reward function, and transition dynamics to ensure the Markov property. Additionally, I reviewed Task 1 to maintain consistency between the MILP formulation and the MDP representation. This concludes my contribution for 40%.

Kedhar Karthik Naidus Rejeti: I have validated the solutions developed by my teammates for Task 1 and Task 2 and confirmed their correctness. This contributes to the overall 20% of this part.

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